Recreational Space Accessibility and Health Outcomes Across Victorian LGAs

ETC5543 Research Project

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Abstract

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1 Introduction

1.1 Background and Motivation

Access to parks and recreational spaces has become an important consideration in Melbourne's urban planning. In October 2024, ABC News reported on a proposal by Infrastructure Victoria to open more public school grounds for community use outside school hours. The proposal was estimated to improve access to nearby recreational spaces for approximately 450,000 Melbourne residents (Jonscher 2024). This initiative reflects efforts to make better use of existing public spaces and improve recreational opportunities for local communities.

Parks, natural areas, and sport and recreation facilities provide spaces for physical activity, outdoor recreation, and social interaction. These activities can contribute to people's physical and mental wellbeing. However, recreational opportunities may vary across communities depending on their local environments and the types of facilities available.

Geographical accessibility is an important aspect of recreational opportunities. It concerns the spatial relationship between where people live and the recreational spaces they may wish to use. Previous research has used distance to the nearest park as one way of measuring accessibility (Shoari et al. 2020). This approach provides a way to examine how proximity to recreational spaces differs between residential locations.

However, greater geographical accessibility to recreational spaces does not necessarily mean higher levels of physical activity or better health outcomes. Factors such as population density and socioeconomic disadvantage may be associated with both recreational accessibility and health outcomes. Examining these relationships while considering such factors is therefore important for understanding the associations between recreational accessibility and population health.

1.2 Research Question and Objectives

The primary research question of this study is:

Is greater accessibility to recreational spaces associated with higher levels of physical activity and better health outcomes across Victorian Local Government Areas (LGAs)?

To address this question, the study has four main objectives:

1. **Measure recreational accessibility:** Develop LGA-level indicators of geographical accessibility to different types of recreational spaces, including parks, natural areas, and sport and recreation

facilities.

2. **Explore geographical patterns:** Examine variations in recreational accessibility, physical activity, and health outcomes across Victorian LGAs.
3. **Investigate statistical associations:** Assess the relationships between recreational accessibility and physical activity and health outcomes, including obesity, psychological distress, and poor self-rated health.
4. **Assess additional explanatory value:** Determine whether recreational accessibility explains additional variation in physical activity and health outcomes after accounting for socioeconomic disadvantage and population density.

1.3 Significance and Contributions

This study provides an LGA-level perspective on recreational space accessibility and its relationship with physical activity and health outcomes across Victoria. By examining different types of recreational spaces, the study offers a broader understanding of how geographical accessibility varies across communities and how these differences relate to population health.

A key contribution of this project is the integration of recreational space, population, socioeconomic, and health data to construct a consistent dataset for analysis across Victorian LGAs. Using population-based spatial locations, the study develops accessibility measures that reflect the geographical proximity of residents to different recreational spaces. This provides a more population-focused perspective on accessibility across LGAs.

The study also contributes empirical evidence on the associations between recreational accessibility and multiple health outcomes, while considering socioeconomic disadvantage and population density. These findings may provide useful insights for future research and discussions about recreational space planning and public health, while recognising that the observed associations do not establish causal relationships.

2 Data and Preparation

2.1 Study Area and Unit of Analysis

This study focuses on Victoria, Australia, using LGAs as the geographical unit of analysis. The LGA level was selected because the physical activity and health outcome indicators are available at this geographical scale, allowing recreational accessibility and other explanatory variables to be examined alongside these outcomes.

As the datasets used in this study have different geographical structures and levels of aggregation, spatial data processing was required to establish a consistent LGA-level framework. The final analytical dataset includes 79 Victorian LGAs, with each LGA representing one observation.

2.2 Data Sources

This study integrates spatial, recreational facility, population, socioeconomic, and health data from multiple sources. These datasets provide information on the distribution of recreational spaces, population characteristics, and health outcomes across Victorian LGAs. The following sections describe the main datasets used in the analysis.

2.2.1 LGA Boundaries

LGA boundary data were obtained from the *Vicmap Admin – Local Government Area (LGA) Polygon Aligned to Property* dataset, provided by the Victorian Government (Department of Transport and Planning 2026a). The dataset contains geographical boundaries of LGAs across Victoria.

These boundaries were used as the common geographical framework for the study, allowing datasets with different spatial structures and levels of aggregation to be integrated at the LGA level.

2.2.2 Recreational Parks and Natural Areas

Spatial data on recreational parks and natural areas were obtained from the Vicmap Planning – Planning Scheme Zone Polygon dataset, provided by the Victorian Government's Department of Transport and Planning (Department of Transport and Planning 2026b). The dataset contains geographical boundaries of planning zones across Victoria.

Two planning zone categories were selected: Public Park and Recreation Zone (PPRZ) and Public Conservation and Resource Zone (PCRZ). PPRZ was used to represent recreational parks, while PCRZ was

used to represent natural areas. These spatial boundaries provide the basis for measuring geographical accessibility to parks and natural areas, as described in Section 3.

2.2.3 Sport and Recreation Facilities

Data on sport and recreation facilities were obtained from the *Sport and Recreational Facilities List*, published by the Victorian Government (Department of Jobs, Skills, Industry and Regions 2024). The dataset contains information on individual recreational facilities across Victoria, including their geographical coordinates, facility types, and sports played.

These data were used to identify the geographical locations of sport and recreation facilities, including indoor facilities. The facility locations provide the basis for measuring geographical accessibility to these recreational spaces, as described in Section 3.

2.2.4 Physical Activity and Health Outcomes

Data on physical activity and health outcomes were obtained from the Victorian Population Health Survey (VPHS) 2023, published by the Victorian Agency for Health Information (Victorian Agency for Health Information 2023). The survey provides population health estimates at the LGA level.

Five indicators were selected for this study: physical inactivity, physical activity (≥150 minutes per week), obesity (BMI ≥30), high or very high psychological distress (K10), and fair or poor self-rated health. These indicators cover physical activity, physical health, and mental wellbeing.

Age-adjusted percentages for the total population (People) were used to support comparisons across LGAs. These indicators serve as the outcome variables for examining their associations with recreational accessibility.

2.2.5 Population and Socioeconomic Data

Population data were obtained from the Australian Bureau of Statistics (ABS). The 2021 Mesh Block boundaries (Australian Bureau of Statistics 2021) and Census Mesh Block Counts (Australian Bureau of Statistics 2022) were used to represent the spatial distribution of residents across Victoria. These data provided the residential locations and population weights used in the accessibility analysis.

LGA-level population estimates for 2023 were obtained from the ABS Regional Population dataset (Australian Bureau of Statistics 2024). These data provided information on population size and density across Victorian LGAs. Population density was included as a control variable in the regression analysis.

Socioeconomic data were obtained from the ABS Socio-Economic Indexes for Areas (SEIFA) 2021 dataset (Australian Bureau of Statistics 2023). The Index of Relative Socio-economic Advantage and Disadvantage (IRSAD) was selected to represent socioeconomic conditions at the LGA level. Higher IRSAD scores indicate relatively greater socioeconomic advantage, while lower scores indicate relatively greater socioeconomic disadvantage. IRSAD was also included as a control variable in the regression analysis.

3 Measuring Recreational Accessibility

4 Statistical Methods

4.1 Outcomes and Explanatory Variables

The analysis examined five outcomes across Victorian Local Government Areas (LGAs), including less than moderate physical activity (inactivity), meeting the recommendation of at least 150 minutes of moderate-to-vigorous physical activity (MVPA) per week, obesity, psychological distress, and fair or poor self-rated health. Each outcome was measured as a percentage at the LGA level.

The explanatory variables comprised socioeconomic conditions, population density, and six recreational accessibility indicators. Socioeconomic conditions were measured using the Socio-Economic Indexes for Areas (SEIFA) Index of Relative Socio-economic Advantage and Disadvantage (IRSAD). Population density was included to account for differences in settlement patterns across LGAs. The accessibility indicators included median distance to parks, natural areas, recreational facilities, and indoor facilities, the percentage of facilities that were indoor facilities, and the number of sport types available.

The analysis assessed whether these accessibility indicators provided additional explanatory power beyond socioeconomic conditions and population density.

4.2 Predictor Transformations

Predictor transformations were applied to reduce skewness. Population density and the four distance measures were log-transformed before standardisation. The percentage of indoor facilities was square-root transformed, while SEIFA IRSAD and sport-type count were retained on their original scales before standardisation. All predictors were standardised using z-scores, calculated by subtracting the sample mean and dividing by the sample standard deviation. The indoor-facility share and sport-type count were reverse-coded so that higher values represented a lower indoor-facility share and fewer sport types. SEIFA IRSAD was not reverse-coded, meaning that higher values continued to indicate greater socioeconomic

advantage. A park–facility composite was constructed as the arithmetic mean of the standardised park-distance and facility-distance indicators. This composite was used in Model A to represent these two dimensions jointly.

4.3 Hierarchical Model Specification

Ordinary least squares (OLS) regression models were fitted separately for each of the five outcomes. A sequence of nested and extended models was used to evaluate the explanatory contribution of socioeconomic conditions, population density, and recreational accessibility.

Table 1: Specification of the hierarchical OLS regression models evaluating socioeconomic conditions, population density, and recreational accessibility.

Model	Predictors	Purpose
M0	SEIFA	Establish the socioeconomic baseline
M1	SEIFA + log-transformed population density	Assess the additional contribution of population density
M2	M1 + one accessibility indicator	Evaluate each accessibility indicator individually
A	M1 + park–facility composite + natural-area distance + indoor-facility distance	Evaluate a compact combination of accessibility indicators
B	M1 + park distance + facility distance + natural-area distance + indoor-facility distance	Evaluate the four distance indicators jointly
C	M1 + all six accessibility indicators	Evaluate the full set of accessibility indicators jointly

For each outcome, M0 and M1 were compared to assess whether population density provided additional explanatory power beyond SEIFA IRSAD. Each M2 model added one accessibility indicator to M1, allowing its individual contribution to be assessed. Models A, B and C examined the joint contribution of multiple accessibility indicators using alternative predictor combinations. The extended models were compared with M1 to evaluate whether recreational accessibility improved explanatory power after accounting for socioeconomic conditions and population density.

4.4 Regional Sensitivity Analysis

To examine whether metropolitan-rural differences influenced the main findings, an additional regional sensitivity analysis was conducted. A binary regional indicator was constructed using the regional classification provided in the VPHS data, with Metropolitan LGAs coded as 1 and Rural LGAs coded as 0 (reference group).

The regional indicator was first added to the baseline model (M1), which included SEIFA IRSAD and log-transformed population density, to create a regional baseline model (M1R). The originally selected accessibility specifications were then retained and extended by adding the same regional indicator, resulting in Models AR, BR and CR. These models were compared with M1R to assess whether the additional explanatory contribution of recreational accessibility remained after accounting for regional differences.

HC3 robust tests were also used to assess the statistical association between the regional indicator and each health outcome, conditional on SEIFA IRSAD and population density. The joint HC3 tests for accessibility predictors were conducted separately to evaluate their additional contribution after regional adjustment.

4.5 Model Comparison and Statistical Inference

Model performance was primarily evaluated using adjusted R^2, which measures the proportion of variation explained by a model while accounting for the number of predictors. For each extended model, the change in adjusted R^2 was calculated relative to M1:

$$R^2_{\text{adj,extended}} - R^2_{\text{adj,M1}}$$

A positive change indicates an increase in adjusted explanatory power, whereas a negative change indicates that the extended model did not improve adjusted R^2 after accounting for model complexity.

Statistical evidence for incremental contributions was assessed using HC3 heteroskedasticity-robust standard errors and robust F-tests. For M1, the test evaluated the contribution of population density beyond SEIFA IRSAD. For each M2 model, it evaluated the individual accessibility indicator added to M1. For Models A, B and C, joint F-tests assessed whether the accessibility predictors added to M1 were jointly different from zero. In the regional sensitivity analysis, the HC3 test for the regional indicator assessed its individual contribution to each baseline model, while joint HC3 tests assessed the additional contribution of accessibility predictors after regional adjustment.

Adjusted (R^2) and HC3 robust tests were interpreted as complementary measures. Adjusted R^2 was

used to compare explanatory power, while the robust tests assessed statistical evidence for the additional predictors. Statistical significance was evaluated at the 5% level. These analyses were conducted at the LGA level and were intended to assess associations and explanatory power rather than establish causal effects.

5 Results

5.1 Baseline Model Performance: M0 vs M1

Model M0 used SEIFA as the sole predictor, while Model M1 additionally included population density. The comparison assessed the explanatory power of socioeconomic conditions and whether population density provided additional explanatory power beyond socioeconomic conditions.

Table 2: Baseline model performance across five outcomes.

Outcome	M0 Adjusted R^2	M1 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	0.617	0.612	-0.005	0.795
At least 150 min/week MVPA	0.163	0.161	-0.002	0.425
Obesity	0.626	0.645	+0.019	0.121
Psychological distress	0.272	0.317	+0.046	0.029
Fair/poor self-rated health	0.437	0.465	+0.027	0.147

Note: Δ Adjusted R^2 is calculated as the adjusted R^2 of M1 minus that of M0. HC3 p-values test the additional contribution of population density after controlling for SEIFA.

The explanatory power of SEIFA varied substantially across outcomes. Under M0, adjusted R^2 ranged from 0.163 for MVPA to 0.626 for obesity, indicating that socioeconomic conditions explained different proportions of the variation across health and physical activity outcomes.

Adding population density produced mixed results. As shown in Table 2,adjusted R^2 increased for obesity, psychological distress and fair/poor self-rated health, with the largest increase observed for psychological distress (+0.046). However, only psychological distress showed statistically significant evidence of an additional contribution from population density at the 5% level (HC3 p = 0.029). The increases for obesity and fair/poor self-rated health were not statistically significant. For inactivity and MVPA, adjusted R^2 decreased slightly after population density was added.

Overall, population density provided additional explanatory power for psychological distress, but its contribution was not consistently supported across the five outcomes. These results establish M1 as the baseline for evaluating whether recreational accessibility provides further explanatory power.

5.2 Individual Accessibility Contributions: M1 to M2

To examine the individual contribution of recreational accessibility, six M2 models were fitted for each outcome, with each model adding one accessibility indicator to M1. Changes in adjusted R^2 and HC3 robust tests were used to evaluate the incremental explanatory contribution of each indicator.

Table 3: Strongest individual accessibility contributions by outcome. *Note: For each outcome, the indicator shown is the one associated with the highest adjusted R^2 among the six individual M2 models. Δ Adjusted R^2 is calculated relative to M1. The complete results for all 30 M2 models are provided in Appendix A.

Outcome	Indicator added	M2 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	Natural-area distance	0.680	+0.068	0.001
At least 150 min/week MVPA	Indoor share (reversed)	0.206	+0.045	0.357
Obesity	Natural-area distance	0.652	+0.008	0.121
Psychological distress	Indoor share (reversed)	0.345	+0.028	0.125
Fair/poor self-rated health	Natural-area distance	0.470	+0.005	0.165

The contribution of individual accessibility indicators differed across outcomes. The clearest improvement was observed for inactivity. Adding natural-area distance increased adjusted R^2 from 0.612 under M1 to 0.680, with statistically significant evidence of an additional contribution (HC3 p = 0.001). The reversed indoor-facility share also improved the adjusted R^2 for inactivity to 0.662 (Δ Adjusted R^2 = 0.050; HC3 p = 0.039). Therefore, both indicators provided evidence of additional explanatory power for inactivity when considered individually. For the other four outcomes, the indicators associated with the highest adjusted R^2 produced smaller improvements, and none had a statistically significant HC3 p-value at the 5% level. For MVPA, the reversed indoor-facility share yielded the largest increase in adjusted R^2 (+0.045), but the evidence was not statistically significant (p = 0.357). Similarly, the largest individual improvements for obesity, psychological distress, and fair/poor self-rated health were not statistically significant.

These findings suggest that individual accessibility indicators provided more substantial additional explanatory power for inactivity than for the other outcomes. However, the results varied across outcomes, and the strongest-performing indicator differed between them. The next section therefore examines whether combining accessibility indicators provides further explanatory power beyond M1.

5.3 Joint Accessibility Contributions: Models A, B and C

To examine whether recreational accessibility indicators provided additional explanatory power when considered jointly, Models A, B and C were compared with the baseline model M1. Each model included SEIFA and population density, while the accessibility indicators were added in different combinations.

Table 4: Joint contribution of accessibility indicators: Models A, B and C versus M1

Outcome	Model	Adjusted R^2	Δ Adjusted R^2	HC3 joint p-value
Less than moderate activity	M1	0.612	–	–
	A	0.713	+0.100	<0.001
	B	0.709	+0.097	<0.001
	C	0.712	+0.100	<0.001
At least 150 min/week MVPA	M1	0.161	–	–
	A	0.158	-0.003	0.795

Outcome	Model	Adjusted R^2	Δ Adjusted R^2	HC3 joint p-value
	B	0.155	-0.005	0.666
	C	0.180	+0.020	0.798
Obesity	M1	0.645	–	–
	A	0.644	-0.001	0.479
	B	0.661	+0.016	0.195
	C	0.653	+0.009	0.367
Psychological distress	M1	0.317	–	–
	A	0.336	+0.018	0.347
	B	0.328	+0.010	0.536
	C	0.336	+0.019	0.370
Fair/poor self-rated health	M1	0.465	–	–
	A	0.461	-0.004	0.281
	B	0.454	-0.011	0.411
	C	0.459	-0.005	0.337

Note: Δ Adjusted R^2 is calculated relative to M1. HC3 joint p-values test whether the accessibility predictors added to each model are jointly different from zero. Values reported as <0.001 are based on the unrounded p-values.

The results showed a clear improvement for less than moderate physical activity. All three combined models increased adjusted R^2 from 0.612 to approximately 0.71. The HC3 joint tests were statistically significant for all three models ($p < 0.001$), indicating that the accessibility indicators provided additional explanatory information beyond socioeconomic conditions and population density. For the other four outcomes, the improvements were smaller and were not statistically significant. Model C produced the highest adjusted R^2 for MVPA (0.180), while Model B performed best for obesity (0.661). For psychological distress, Models A and C achieved similar adjusted R^2 values of 0.336. However, none of these models showed a significant joint contribution from accessibility indicators. For fair or poor self-rated health, all three combined models had lower adjusted R^2 values than M1.

To further compare the three combinations, nested model comparisons were conducted between Models A and B, and between Models B and C.

Table 5: Nested comparisons among combined models.

Outcome	Comparison	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	A → B	-0.003	0.675
	B → C	+0.003	0.567
At least 150 min/week MVPA	A → B	-0.002	0.384
	B → C	+0.025	0.607
Obesity	A → B	+0.017	0.071
	B → C	-0.008	0.797
Psychological distress	A → B	-0.008	0.701
	B → C	+0.008	0.216
Fair/poor self-rated health	A → B	-0.007	0.864
	B → C	+0.006	0.222

Note: Δ Adjusted R^2 represents the change in adjusted R^2 when moving from the first model to the second. HC3 p-values assess the additional contribution of the predictors introduced in the larger model.

The nested comparisons did not show statistically significant improvements when moving from Model A to B or from Model B to C at the 5% significance level. Although adding the separate park and facility distance indicators in Model B increased adjusted R^2 for obesity by 0.017 compared with Model A, this improvement was not statistically significant ($p = 0.071$). Similarly, adding the remaining accessibility indicators in Model C did not significantly improve any outcome.

Overall, the combined results suggest that the explanatory contribution of recreational accessibility varied across health outcomes. The strongest and most consistent evidence was found for inactivity, while the results for the other outcomes provided limited statistical evidence of additional explanatory power.

5.4 Comparison of Accessibility Dimensions

The individual M2 models and combined accessibility models provide complementary evidence about the contribution of different accessibility dimensions. The strongest individual indicator and the best-performing combined model varied across health outcomes, as summarised in Table Table 6. The coefficient estimates and their HC3 robust confidence intervals for the selected models are provided in Appendix B. These results offer additional information about the direction and uncertainty of individual predictor associations.

Table 6: Summary of the strongest individual and combined accessibility models.

Outcome	Strongest individual indicator (M2)	Best combined model	Interpretation
Less than moderate activity	Natural-area distance	Model A	Clear improvement from accessibility indicators, supported by significant individual and joint tests
At least 150 min/week MVPA	Indoor-facility share (reversed)	Model C	Adjusted R^2 increased slightly, but the joint contribution was not significant
Obesity	Natural-area distance	Model B	Some improvement in adjusted R^2, but no significant joint contribution
Psychological distress	Indoor-facility share (reversed)	Model A	Small improvement in adjusted R^2, without significant statistical evidence
Fair/poor self-rated health	Natural-area distance	M1	Accessibility indicators did not improve the adjusted R^2

Overall, natural-area distance showed the largest individual contribution for inactivity, obesity and fair or poor self-rated health, while the reversed indoor-facility share showed the largest individual contribution for MVPA and psychological distress. However, only inactivity showed clear statistical evidence that accessibility indicators jointly added explanatory power. These findings suggest that the contribution of recreational accessibility varies across health outcomes rather than showing a consistent pattern across all health measures.

6 Model Diagnostics and Robustness

Several diagnostic and robustness checks were conducted to assess the reliability of the regression results. These included checks for multicollinearity, heteroskedasticity and influential observations, a leave-one-LGA-out analysis, and tests for spatial autocorrelation in model residuals.

6.1 Regression diagnostics

Residual diagnostic plots for the three inactivity models are provided in Appendix C. The plots showed some departures from normality, which were considered alongside the numerical diagnostic tests and robustness analysis.

Variance Inflation Factors (VIF) were used to assess multicollinearity, while the Breusch–Pagan test was used to examine heteroskedasticity. Cook's distance was used to identify potentially influential observations. The diagnostic results for the selected model of each outcome are summarised below. Complete diagnostic results for all candidate models are reported in Appendix D.

Table 7: Diagnostic results for selected models.

Outcome	Model	Max VIF	BP p-value	Max Cook's distance	Observations with Cook's D > 4/n
Less than moderate activity	A	3.981	0.358	0.121	5
At least 150 min/week MVPA	C	5.959	0.077	0.671	7
Obesity	B	3.981	0.132	0.230	5
Psychological distress	A	3.981	0.092	0.176	6
Fair/poor self-rated health	M1	2.061	0.007	0.559	2

Note: An observation was flagged when Cook's distance exceeded (4/n), where (n = 79). These flags

indicate potentially influential observations and do not automatically imply that the observations should be removed.

As shown in Table 7, the maximum VIF values ranged from 2.061 to 5.959. Model C for MVPA had the highest value, suggesting greater overlap among its predictors compared with the other selected models. However, the VIF results did not indicate extremely high multicollinearity.

The Breusch–Pagan test identified potential heteroskedasticity in the selected M1 model for fair or poor self-rated health ($p = 0.007$). The tests for the other four selected models were not statistically significant at the 5% level. Therefore, HC3 robust standard errors were used throughout the analysis to provide more reliable statistical inference when heteroskedasticity was present.

Cook's distance identified several potentially influential observations. The largest maximum value was observed in the MVPA model (0.671), followed by the fair or poor self-rated health model (0.559). These observations were retained in the analysis, but their potential influence was further examined through the LOO analysis.

6.2 Leave-one-LGA-out analysis

A leave-one-LGA-out (LOO) analysis was conducted to examine whether the regression results were overly dependent on any single LGA. This analysis focused on the stability of model fit and the estimated coefficient for natural-area distance (`z_nature`). Natural-area distance was selected for this check because it showed the largest individual improvement in adjusted R^2 for inactivity and was included in all three combined accessibility models (A, B and C). The aim was to assess whether its estimated association remained consistent when individual LGAs were excluded, rather than to examine the stability of every accessibility coefficient. Each candidate combined model was refitted 79 times, with one LGA excluded in each iteration. The minimum and maximum adjusted R^2 values and the corresponding range of the natural-area distance coefficient were recorded.

Table 8: Leave-one-LGA-out results for selected combined accessibility models

Outcome	Model	Min. adjusted R^2	Max. adjusted R^2	Min. natural-area coefficient	Max. natural-area coefficient
Less than moderate activity	A	0.688	0.735	1.000	1.250
At least 150 min/week MVPA	C	0.093	0.212	-0.791	-0.239
Obesity	B	0.641	0.697	0.710	1.088

Outcome	Model	Min. adjusted R^2	Max. adjusted R^2	Min. natural-area coefficient	Max. natural-area coefficient
Psychological distress	A	0.308	0.368	-0.515	-0.134

Note: The main-text table reports the selected combined model for each outcome where the selected model included natural-area distance. The selected M1 model for fair/poor self-rated health is not shown because it does not contain accessibility indicators. However, Appendix E includes Models A, B and C for this outcome.

For less than moderate physical activity, the adjusted R^2 of the selected Model A ranged from 0.688 to 0.735, remaining above the M1 baseline of 0.612. The coefficient for natural-area distance also remained positive across all 79 refitted models. This suggests that the improvement in explanatory power was not driven entirely by a single LGA. For the other outcomes, the adjusted R^2 values also varied across the LOO models. However, the estimated coefficient for natural-area distance retained the same direction within each outcome: negative for MVPA and psychological distress, and positive for obesity. This indicates that the direction of this estimated association was stable, although its magnitude and overall model fit varied depending on which LGA was excluded. Complete LOO results for all 15 combined accessibility models are reported in Appendix E.

Overall, the LOO analysis provides the strongest robustness support for the inactivity result. However, this check only examines the stability of the natural-area distance coefficient and does not establish the stability of other accessibility coefficients or their statistical significance. The results for outcomes other than inactivity should therefore be interpreted more cautiously.

6.3 Residual spatial autocorrelation

Moran's I tests were conducted to examine whether residuals from geographically neighbouring LGAs were spatially correlated. The tests were applied to the selected model for each outcome using a queen-contiguity spatial weights matrix.

Table 9: Moran's I tests for residual spatial autocorrelation.

Outcome	Model	Moran's I	p-value	Result
Less than moderate activity	A	-0.012	0.994	Not significant

Outcome	Model	Moran's I	p-value	Result
At least 150 min/week MVPA	C	0.144	0.028	Significant positive spatial autocorrelation
Obesity	B	-0.083	0.323	Not significant
Psychological distress	A	-0.020	0.919	Not significant
Fair/poor self-rated health	M1	-0.095	0.245	Not significant

The results identified statistically significant positive spatial autocorrelation in the residuals of the MVPA model (Model C; Moran's I = 0.144, p = 0.028). This suggests that geographically neighbouring LGAs may have similar unexplained variation in MVPA, indicating that the model may not fully capture spatial patterns in physical activity. In Table 9, no statistically significant spatial autocorrelation was detected in the residuals of the other four selected models. In particular, the inactivity model showed no evidence of residual spatial autocorrelation (Moran's I = -0.012, p = 0.994). However, a non-significant result does not prove that all spatial dependence has been eliminated.

Overall, the spatial diagnostic results suggest that the MVPA model requires particular caution, while no significant residual spatial autocorrelation was detected in the other selected models.

Taken together, the diagnostic and robustness checks provide the strongest support for the stability of the inactivity result. Its improvement in adjusted R^2 remained above the baseline level across the LOO analysis, and no significant residual spatial autocorrelation was detected. However, the MVPA model showed significant spatial autocorrelation, while the other outcomes had more limited improvements in explanatory power. These limitations should be considered when interpreting the regression findings.

6.4 Additional sensitivity analysis: Regional differences

To examine whether the main findings were influenced by differences between metropolitan and rural areas, an additional sensitivity analysis was conducted using the regional classification provided in the VPHS data. A binary regional indicator was added to the baseline models, distinguishing Metropolitan LGAs from Rural LGAs. This allowed us to examine whether the additional explanatory contribution of recreational accessibility remained after accounting for socioeconomic conditions, population density and regional differences.

Table 10: Sensitivity analysis: Metropolitan–Rural regional adjustment and accessibility contributions

Outcome	Original selected model	Original selected Adj. R^2	Original ΔAdj. R^2	Original HC3 joint p	Region-adjusted selected model	M1R Adj. R^2	Region-adjusted selected Adj. R^2	Region-adjusted ΔAdj. R^2	Region-adjusted HC3 joint p
Less than moderate activity	A	0.713	0.100	<0.001	AR	0.651	0.737	0.086	<0.001
At least 150 min/week MVPA	C	0.180	0.020	0.798	CR	0.278	0.307	0.030	0.352
Obesity	B	0.661	0.016	0.195	BR	0.640	0.657	0.017	0.194
Psychological distress	A	0.336	0.018	0.347	AR	0.323	0.337	0.014	0.362
Fair/poor self-rated health	M1	0.465	0.000	—	M1R	0.459	0.459	0.000	—

Note: M1R includes SEIFA, log population density and a regional indicator (Metropolitan = 1; Rural = 0). Models AR, BR and CR extend M1R using the same accessibility specifications as the originally selected Models A, B and C, respectively. Original ΔAdjusted R^2 represents the improvement over M1 before regional adjustment, while Regional ΔAdjusted R^2 represents the improvement over M1R. HC3 joint p-values test the joint contribution of the added accessibility indicators. No accessibility indicators were added for fair/poor self-rated health, as M1R was retained as the selected model.

The regional indicator was statistically significant in the baseline models for inactivity ($p = 0.0166$) and MVPA ($p = 0.0084$), after controlling for SEIFA and population density. However, it was not statistically significant for obesity, psychological distress or fair/poor self-rated health. This suggests that metropolitan-rural differences may be relevant to explaining variation in physical activity outcomes,

although their relevance varies across health outcomes. The full HC3 test results are provided in Appendix F. Table 10 showed that the additional explanatory contribution of recreational accessibility for inactivity remained statistically significant after regional adjustment. Adjusted R^2 increased from 0.651 for the regional baseline model (M1R) to 0.737 for Model AR, with a significant HC3 robust joint test ($p < 0.001$). Although the improvement was smaller than in the original model, accessibility continued to provide substantial additional explanatory power. For MVPA, obesity and psychological distress, accessibility produced only small additional improvements in adjusted R^2, and none of the corresponding HC3 robust joint tests was statistically significant. No improvement was observed for fair/poor self-rated health.

Overall, the regional sensitivity analysis supports the robustness of the main finding for inactivity. The additional explanatory contribution of accessibility remained significant after accounting for the metropolitan-rural classification, suggesting that the finding was not entirely explained by broad regional differences.

7 Discussion

7.1 RQ1: The explanatory role of socioeconomic conditions and population density

The baseline models showed that socioeconomic advantage was an important explanatory variable, although its explanatory power varied across health outcomes. The adjusted R^2 of the SEIFA-only models ranged from 0.163 for achieving at least 150 minutes of moderate-to-vigorous physical activity per week to 0.626 for obesity. This suggests that socioeconomic conditions were more strongly associated with some health outcomes than others.

Adding population density produced limited improvements for most outcomes. The clearest improvement was observed for psychological distress, for which adjusted R^2 increased from 0.272 to 0.317. In contrast, the addition of population density did not improve adjusted R^2 for inactivity or MVPA. Therefore, population density provided outcome-specific rather than consistent additional explanatory value beyond socioeconomic advantage.

The regional sensitivity analysis further examined whether metropolitan-rural differences were associated with health outcomes. After controlling for SEIFA and population density, the regional indicator was statistically significant for inactivity ($p = 0.0166$) and MVPA ($p = 0.0084$), but not for obesity, psychological distress or fair/poor self-rated health. The adjusted R² of the baseline model also increased

from 0.161 to 0.278 for MVPA after adding the regional indicator. These results suggest that broad geographic differences may be relevant to explaining variation in physical activity outcomes, particularly MVPA.

7.2 RQ2: Does recreational accessibility improve explanatory power?

The contribution of recreational accessibility differed substantially across outcomes. The strongest evidence was found for inactivity. Adding accessibility indicators in Model A increased adjusted R^2 from 0.612 to 0.713, and the HC3 robust joint test was statistically significant ($p < 0.001$). This indicates that accessibility measures jointly improved the explanation of differences in inactivity across LGAs, beyond socioeconomic advantage and population density. For MVPA, obesity and psychological distress, the selected models produced only modest increases in adjusted R^2, and none of the corresponding HC3 robust joint tests was statistically significant. For fair/poor self-rated health, the accessibility models did not improve adjusted R^2 relative to the baseline model. Overall, the results do not support a consistent additional contribution of recreational accessibility across all health outcomes.

The regional sensitivity analysis provided further evidence of the robustness of these findings. For inactivity, the additional adjusted R^2 from accessibility decreased from 0.100 in the original model to 0.086 after regional adjustment. However, the HC3 robust joint test remained statistically significant ($p < 0.001$). This suggests that the additional explanatory contribution of accessibility for inactivity was not entirely explained by metropolitan-rural differences. For MVPA, obesity and psychological distress, accessibility continued to provide only modest improvements in adjusted R^2, with no statistically significant joint contributions. No improvement was observed for fair/poor self-rated health.

The distinction between inactivity and MVPA is noteworthy. Although these outcomes are related to physical activity, they measure different aspects of activity behaviour. The results suggest that area-level recreational accessibility may be more informative for explaining variation in inactivity than variation in meeting the weekly MVPA threshold. Regional differences may also help explain some of the variation in MVPA. However, the observational design does not establish why these differences occur.

7.3 RQ3: Which accessibility dimensions contribute most?

The individual-indicator analyses suggested that natural-area distance and the indoor-facility share indicator were particularly relevant to inactivity. In the combined models, accessibility measures also provided a substantial overall improvement for this outcome. This suggests that the relationship may involve multiple aspects of recreational provision rather than one isolated measure. However, the results do not establish that one accessibility dimension is universally the most important. The combined models did

not show statistically significant joint contributions for most of the other health outcomes. In addition, predictors may overlap in the information they capture, particularly in Model C, which had a higher maximum VIF than Models A and B. Individual coefficients should therefore be interpreted alongside their confidence intervals, the joint tests and the model diagnostics.

7.4 Overall implications

Taken together, the findings suggest that the explanatory value of recreational accessibility depends on the health outcome being considered. Accessibility measures provided the clearest additional explanatory value for inactivity, and this finding remained after controlling for metropolitan-rural differences. The regional sensitivity analysis also suggested that geographic context may be relevant to explaining variation in MVPA. These findings support evaluating specific health outcomes separately rather than assuming that greater recreational accessibility is associated with uniformly better health across all measures. The results are area-level associations and should not be interpreted as evidence that changing recreational accessibility will necessarily cause changes in individual behaviour or health. The significant residual spatial autocorrelation in the MVPA model also indicates that its results require particular caution.

8 Limitations

This study has several limitations. First, the analysis is conducted at the LGA level, so the results describe area-level associations and should not be interpreted as individual-level relationships. Second, the cross-sectional nature of the data limits causal interpretation. The findings indicate associations rather than causal effects. Third, the accessibility measures are proxies for recreational provision and do not fully capture actual travel time, facility quality, affordability, opening hours or individual preferences. Fourth, the regional sensitivity analysis used a binary classification distinguishing Metropolitan from Rural LGAs, based on the regional classification in the VPHS data. Although this helped account for broad metropolitan-rural differences, it may not fully capture variation within each group or differences between urban, peri-urban and rural communities. Therefore, some geographic differences may remain unaccounted for. Finally, the residuals of the selected MVPA model showed statistically significant spatial autocorrelation. This suggests that the model may not fully account for geographic patterns in physical activity. Although HC3 robust inference was used to address heteroskedasticity, it does not resolve spatial dependence. The MVPA findings should therefore be interpreted cautiously, and future work could investigate spatial regression methods or additional geographic and environmental variables.

Despite these limitations, the analysis provides evidence that recreational accessibility may offer addi-

tional explanatory value for inactivity at the LGA level. The regional sensitivity analysis further suggests that this finding is not entirely explained by broad metropolitan-rural differences. The strength of these relationships differs across health outcomes, highlighting the importance of evaluating each outcome separately.

9 Conclusion

This study examined whether recreational space accessibility helps explain differences in physical activity and health outcomes across Victorian LGAs, after accounting for socioeconomic advantage and population density.

First, socioeconomic advantage explained varying proportions of the differences in health outcomes, while population density provided additional explanatory value mainly for selected outcomes, particularly psychological distress. The supplementary regional analysis showed that metropolitan-rural differences were statistically associated with inactivity and MVPA, with the largest increase in adjusted R^2 observed for MVPA. Second, recreational accessibility provided clear additional explanatory value for inactivity, increasing adjusted R^2 from 0.612 to 0.713 with a statistically significant HC3 robust joint test. This finding remained after controlling for regional differences, with the additional adjusted R^2 decreasing slightly from 0.100 to 0.086. However, comparable evidence was not found for MVPA, obesity, psychological distress or fair/poor self-rated health. Third, natural-area distance and the indoor-facility share indicator showed notable contributions in the individual-indicator analyses for inactivity, although the results do not establish a single universally dominant accessibility dimension.

Overall, recreational accessibility appears to be more informative for explaining differences in inactivity than for explaining all physical activity and health outcomes across Victorian LGAs. These findings highlight the importance of considering both health outcomes and geographic context when examining recreational provision. Further research using more detailed measures of accessibility and methods that account for spatial dependence would help assess the robustness and broader applicability of these associations.

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Appendix

Appendix A. Additional Model Results (M1 vs M2)

Table 11: Incremental explanatory power of accessibility indicators (M1 → M2).

Outcome	Indicator added	M2 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
Less than moderate activity	Park distance	0.619	+0.007	0.138
Less than moderate activity	Natural-area distance	0.680	+0.068	0.001
Less than moderate activity	Facility distance	0.621	+0.009	0.162
Less than moderate activity	Indoor-facility distance	0.630	+0.017	0.071
Less than moderate activity	Indoor share (reversed)	0.662	+0.050	0.039
Less than moderate activity	Sport diversity (reversed)	0.609	-0.004	0.701
At least 150 min/week MVPA	Park distance	0.150	-0.011	0.908
At least 150 min/week MVPA	Natural-area distance	0.175	+0.015	0.300
At least 150 min/week MVPA	Facility distance	0.156	-0.005	0.440
At least 150 min/week MVPA	Indoor-facility distance	0.154	-0.007	0.667
At least 150 min/week MVPA	Indoor share (reversed)	0.206	+0.045	0.357
At least 150 min/week MVPA	Sport diversity (reversed)	0.166	+0.005	0.443
Obesity	Park distance	0.641	-0.003	0.630
Obesity	Natural-area distance	0.652	+0.008	0.121
Obesity	Facility distance	0.645	0.000	0.297
Obesity	Indoor-facility distance	0.641	-0.004	0.698
Obesity	Indoor share (reversed)	0.640	-0.004	0.765
Obesity	Sport diversity (reversed)	0.643	-0.001	0.442
Psychological distress	Park distance	0.315	-0.002	0.347

Outcome	Indicator added	M2 Adjusted R^2	Δ Adjusted R^2	HC3 p-value
Psychological distress	Natural-area distance	0.308	-0.009	0.927
Psychological distress	Facility distance	0.322	+0.005	0.152
Psychological distress	Indoor-facility distance	0.319	+0.002	0.355
Psychological distress	Indoor share (reversed)	0.345	+0.028	0.125
Psychological distress	Sport diversity (reversed)	0.308	-0.009	0.973
Fair/poor self-rated health	Park distance	0.465	+0.001	0.301
Fair/poor self-rated health	Natural-area distance	0.470	+0.005	0.165
Fair/poor self-rated health	Facility distance	0.462	-0.002	0.439
Fair/poor self-rated health	Indoor-facility distance	0.458	-0.007	0.915
Fair/poor self-rated health	Indoor share (reversed)	0.464	-0.001	0.297
Fair/poor self-rated health	Sport diversity (reversed)	0.469	+0.005	0.229

Note: This is the complete results for all 30 M2 models. For each outcome, the indicator shown is the one associated with the highest adjusted R^2 among the six individual M2 models. Δ Adjusted R^2 is calculated relative to M1.

Appendix B. Selected Model Coefficients and HC3 Confidence Intervals

Table 12: Selected Model Coefficients and HC3 Confidence Intervals.

Outcome	Model	Term	Coefficient	HC3_SE	CI_lower	CI_upper	HC3_p
Less than moderate activity	A	(Intercept)	17.695	0.290	17.117	18.272	0.000
Less than moderate activity	A	z_seifa	-3.717	0.442	-4.597	-2.836	0.000
Less than moderate activity	A	z_density	-0.542	0.645	-1.827	0.743	0.403
Less than moderate activity	A	z_parkfac	1.154	0.407	0.343	1.964	0.006
Less than moderate activity	A	z_nature	1.119	0.344	0.433	1.804	0.002
Less than moderate activity	A	z_indoor	-1.290	0.436	-2.159	-0.421	0.004
At least 150 min/week MVPA	C	(Intercept)	35.127	0.467	34.195	36.058	0.000
At least 150 min/week MVPA	C	z_seifa	1.758	0.853	0.056	3.460	0.043
At least 150 min/week MVPA	C	z_density	0.887	1.719	-2.541	4.314	0.607
At least 150 min/week MVPA	C	z_park	0.731	0.778	-0.821	2.282	0.351
At least 150 min/week MVPA	C	z_nature	-0.552	0.590	-1.729	0.626	0.353
At least 150 min/week MVPA	C	z_facility	-0.660	0.836	-2.328	1.007	0.432

Outcome	Model	Term	Coefficient	HC3_SE	CI_lower	CI_upper	HC3_p
At least 150 min/week MVPA	C	z_indoor	-0.501	0.907	-2.310	1.308	0.582
At least 150 min/week MVPA	C	z_indoor_share	1.593	1.615	-1.629	4.815	0.327
At least 150 min/week MVPA	C	z_sport	0.174	0.824	-1.470	1.818	0.833
Obesity	B	(Intercept)	24.969	0.443	24.085	25.852	0.000
Obesity	B	z_seifa	-4.008	0.869	-5.739	-2.276	0.000
Obesity	B	z_density	-1.567	1.153	-3.866	0.732	0.179
Obesity	B	z_park	-1.641	0.973	-3.581	0.300	0.096
Obesity	B	z_facility	1.432	0.884	-0.331	3.194	0.110
Obesity	B	z_nature	0.913	0.485	-0.053	1.879	0.064
Obesity	B	z_indoor	0.431	0.640	-0.845	1.708	0.503
Psychological distress	A	(Intercept)	19.791	0.394	19.006	20.576	0.000
Psychological distress	A	z_seifa	-3.133	0.595	-4.319	-1.947	0.000
Psychological distress	A	z_density	1.601	0.873	-0.140	3.341	0.071
Psychological distress	A	z_parkfac	1.270	0.730	-0.185	2.725	0.086
Psychological distress	A	z_nature	-0.294	0.504	-1.299	0.710	0.561
Psychological distress	A	z_indoor	-1.091	0.769	-2.624	0.443	0.161
Fair/poor self-rated health	M1	(Intercept)	21.672	0.387	20.901	22.442	0.000
Fair/poor self-rated health	M1	z_seifa	-3.899	0.764	-5.420	-2.377	0.000
Fair/poor self-rated health	M1	z_density	1.200	0.819	-0.432	2.831	0.147

Appendix C. Residual Diagnostic Plots for Inactivity Models

Appendix D. Complete Regression Diagnostic Results

Table 13: Complete Regression Diagnostic Results

Outcome	Model	n	adj_r2	delta_adj_r2	max_vif	bp_p	max_cooks	cooks_over_4n
At least 150 min/week MVPA	A	79	0.158	-0.003	3.981	0.016	0.485	9
At least 150 min/week MVPA	B	79	0.155	-0.005	3.981	0.023	0.421	8
At least 150 min/week MVPA	C	79	0.180	0.020	5.959	0.077	0.671	7
At least 150 min/week MVPA	M1	79	0.161	0.000	2.061	0.177	0.164	7
Psychological distress	A	79	0.336	0.018	3.981	0.092	0.176	6
Psychological distress	B	79	0.328	0.010	3.981	0.089	0.154	7
Psychological distress	C	79	0.336	0.019	5.959	0.211	0.306	4
Psychological distress	M1	79	0.317	0.000	2.061	0.434	0.118	4
Less than moderate activity	A	79	0.713	0.100	3.981	0.358	0.121	5
Less than moderate activity	B	79	0.709	0.097	3.981	0.408	0.116	6
Less than moderate activity	C	79	0.712	0.100	5.959	0.448	0.208	6
Less than moderate activity	M1	79	0.612	0.000	2.061	0.749	0.112	3
Obesity	A	79	0.644	0.000	3.981	0.020	0.267	5
Obesity	B	79	0.661	0.016	3.981	0.132	0.230	5
Obesity	C	79	0.653	0.009	5.959	0.136	0.196	6
Obesity	M1	79	0.645	0.000	2.061	0.004	0.496	3
Fair/poor self-rated health	A	79	0.461	-0.004	3.981	0.035	0.315	5
Fair/poor self-rated health	B	79	0.454	-0.011	3.981	0.048	0.308	6

Outcome	Model	n	adj_r2	delta_adj_r2	max_vif	bp_p	max_cooks	cooks_over_4n
Fair/poor self-rated health	C	79	0.459	-0.005	5.959	0.154	0.281	3
Fair/poor self-rated health	M1	79	0.465	0.000	2.061	0.007	0.559	2

Note: The table reports diagnostic results for all 20 candidate models, including M1, A, B and C for each of the five outcomes. VIF measures multicollinearity, while the Breusch–Pagan (BP) test examines heteroskedasticity. Cook's distance is used to identify potentially influential observations. The threshold of 4/n is used as a screening guideline, and observations exceeding this threshold are not automatically removed.

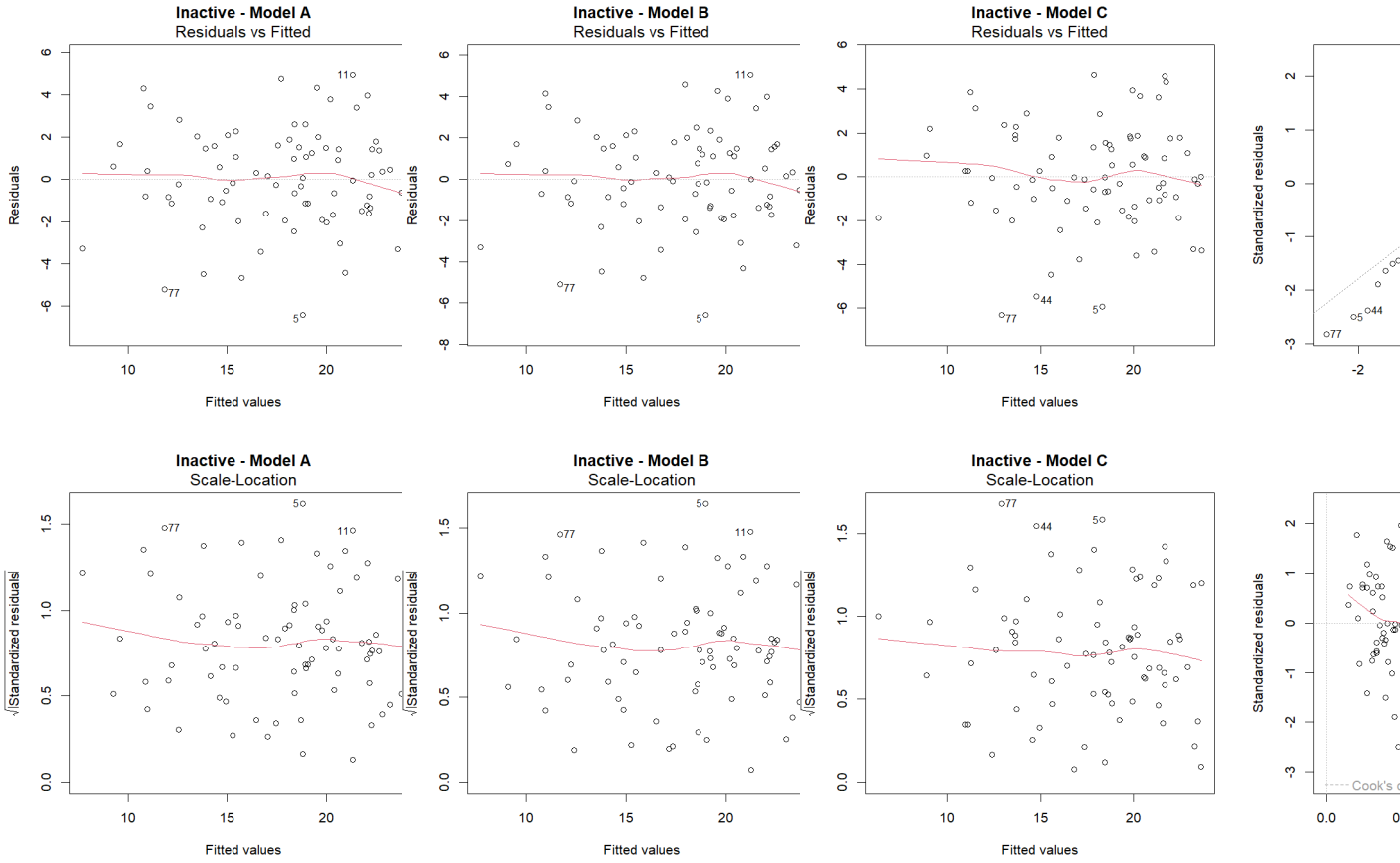
Appendix E. Complete Leave-One-LGA-Out Results

Table 14: Complete Leave-One-LGA-Out Results

outcome	model	min_adj_r2	max_adj_r2	min_nature_coef	max_nature_coef
At least 150 min/week MVPA	A	0.113	0.191	-0.893	-0.229
At least 150 min/week MVPA	B	0.115	0.184	-0.912	-0.267
At least 150 min/week MVPA	C	0.093	0.212	-0.791	-0.239
Psychological distress	A	0.308	0.368	-0.515	-0.134
Psychological distress	B	0.300	0.360	-0.500	-0.120
Psychological distress	C	0.304	0.368	-0.606	-0.235
Less than moderate activity	A	0.688	0.735	1.000	1.250
Less than moderate activity	B	0.684	0.734	1.011	1.263
Less than moderate activity	C	0.682	0.733	0.920	1.181
Obesity	A	0.626	0.687	0.560	1.007
Obesity	B	0.641	0.697	0.710	1.088
Obesity	C	0.632	0.691	0.735	1.102
Fair/poor self-rated health	A	0.436	0.543	0.229	0.676
Fair/poor self-rated health	B	0.429	0.539	0.181	0.669
Fair/poor self-rated health	C	0.436	0.552	0.140	0.594

Note:For each outcome–model combination, the model was refitted 79 times, with one LGA excluded in each iteration. The table reports the minimum and maximum adjusted R^2 values and the corresponding range of the coefficient for natural-area distance (`z_nature`). All Models A, B and C include this predictor. The coefficient ranges assess the stability of the estimated association with natural-area distance only; they do not assess the stability of other accessibility coefficients or statistical significance. Model M1 is not included because it does not contain accessibility indicators.

Appendix F. HC3 Robust Tests for the Regional Indicator



(a) (a) Inactive – Model A (b) (b) Inactive – Model B (c) (c) Inactive – Model C

Figure 1: Residual diagnostic plots for the three inactivity models.

Table 15: HC3 Robust Tests for the Regional Indicator

Health outcome	Regional indicator HC3 p-value	Evidence at 5% level
Less than moderate activity (inactivity)	0.0166	Yes
MVPA (at least 150 min/week)	0.0084	Yes
Obesity	0.9690	No
Psychological distress	0.3290	No
Fair/poor self-rated health	0.6330	No

Note: The regional indicator was included in the M1R models alongside SEIFA and log population density. Metropolitan was coded as 1 and Rural as 0 (reference group). HC3 robust p-values test whether the regional indicator is statistically associated with each health outcome, conditional on the other variables in M1R.